

RMIT International University Vietnam

Assignment Cover Page (INDIVIDUAL)

Course Code:	ECON1612 ECON1613
Course Name:	Big Data, Machine Learning and Society
Location (Campus):	SGS
Assignment Number:	Assignment 1
Title of Assignment:	Causal and Predictive Analysis for EVN
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Due Date:	Friday, 14 th November 2025
Date of Submission:	Friday, 14 th November 2025
Number of pages including this one:	7
Word Count:	1095 words

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Q1. Business vs. Policy Questions: Causal and Predictive

Vietnam Electricity (EVN), the state-owned national utility, is responsible for powering the nation's rapid economic growth. Its core objectives are to ensure a stable electricity supply, maintain financial health, and support energy security. Surging demand puts immense pressure on EVN to improve efficiency and make sound investments, making rigorous data analysis essential for its business performance and policy fulfillment.

a) Causal Questions:

Causal Question 1: What is the causal effect of implementing a time-of-use (TOU) pricing scheme on the peak-hour electricity consumption of industrial customers in Binh Duong province?

Justification: Inferring causality matters for investment decisions. EVN must know if the TOU policy itself causes lower peak demand, distinct from other economic trends.

Causal Question 2: What is the causal effect of increased preventative maintenance expenditure on the frequency of power outages in Ho Chi Minh City's distribution network?

Justification: Causality is vital for budget optimization. Proving that higher spending directly causes fewer outages justifies a significant operational cost.

b) Predictive Questions:

Predictive Question 1: Which residential customers in Da Nang are most likely to default on their electricity bills in the next quarter?

Explanation: Accurate prediction enables proactive risk management. By identifying high-risk customers, EVN can target them with supportive interventions like payment plans instead of reactive collections.

Predictive Question 2: What is the projected electricity demand in the Red River Delta region for each hour of the day over the next seven days?

Explanation: Accurate forecasting is critical for efficient grid operation. It allows EVN to optimize power generation dispatch, prioritizing cheaper, cleaner energy sources.

c) Importance of Solving Each Question:

Causal 1 (TOU Pricing): The independent variable is TOU scheme implementation (yes/no), and the dependent variable is peak-hour electricity consumption (kWh). A confirmed causal link validates the use of pricing policy to reduce peak demand, helping defer expensive power plant construction. This could save billions and enhance national energy security. It directly

informs EVN's investment strategy and supports Vietnam's Power Development Plan 8 (PwC 2025), which prioritizes non-generation solutions for rising demand.

Causal 2 (Maintenance): The independent variable is maintenance level (VND), and the dependent variable is outage frequency (count). Understanding this causal relationship enables EVN to allocate resources more efficiently to improve service reliability. This has direct implications for operational budgeting and public perception. It is especially critical for ensuring stable power supply in HCMC, benefiting EVN operations, consumers, and regulators.

Predictive 1 (Customer Default): The dependent variable is future default status (yes/no), predicted using independent variables like payment history (On-time, Late, Unpaid) and average monthly usage (kWh). Accurate predictions help EVN mitigate revenue loss, reduce collection costs, and prevent service disconnections. It supports financial planning and assists customers facing hardship. In a volatile economy, this approach protects revenue while promoting social responsibility.

Predictive 2 (Demand Forecasting): The dependent variable is hourly electricity demand (MWh), with independent variables including time-series and weather data such as temperature, humidity, and cloud cover. Forecasting demand precisely helps lower operational costs, reduce emissions, and maintain grid stability. It prevents blackouts and supports efficient supply-demand matching. This is essential for managing the growing share of intermittent renewable energy in Vietnam's grid (EVNEIC 2025).

Q2. Methods Choice:

a) Econometric Designs for Causal Questions:

Causal 1 (TOU Pricing): Difference-in-Differences (DiD): This design would compare the change in consumption in Binh Duong (treatment group) before and after the policy against a comparable province without the policy (control group). DiD isolates the policy's causal impact by removing common time trends. The key assumption is parallel trends, both groups would have shown similar consumption patterns if the policy had not been introduced.

Causal 2 (Preventative Maintenance): Panel/Fixed Effects Model: Using data from numerous districts over several years, this method controls unobserved, time-invariant district characteristics like infrastructure age. This isolates the causal link between changes in maintenance spending and changes in outage frequency within each district. The key assumption is that no unobserved time-varying factors influence both maintenance levels and outage rates.

b) Machine Learning Approaches for Predictive Questions:

Predictive 1 (Customer Default): Gradient Boosted Decision Tree (GBDT): This classification model is ideal for handling the mix of numeric and categorical customer data. It builds an ensemble of decision trees to capture complex, non-linear patterns, generating a highly accurate default risk score for each customer. Its suitability comes from its high out-of-sample predictive performance.

Predictive 2 (Demand Forecasting): Long Short-Term Memory (LSTM) Neural Network: As a recurrent neural network, LSTM is designed for time-series data. Its internal memory can learn the complex daily, weekly, and seasonal patterns in historical electricity load and weather data, making it highly effective for producing accurate hourly demand forecasts.

c) Differentiating Econometrics vs. Machine Learning:

The core difference is causation versus prediction. The econometric methods (DiD, Fixed Effects) are used to answer "Why?", to provide a single, interpretable estimate of a policy's causal impact for a strategic "go/no-go" decision. In contrast, the ML models (GBDT, LSTM) answer "What will happen?", generating the most accurate predictions possible from complex data to enable precise operational targeting. The econometric models deliver a transparent explanation, while the ML models deliver high-performance predictive accuracy, even if the model itself is a "black box."

Q3. Data Sources, Feasibility, and Ethics:

a) Concrete Data Sources for Predictive Questions:

For customer default prediction, internal data from EVN's CRM and Billing System includes anonymized customer IDs, monthly usage, payment history, and tenure. This data is not publicly accessible and must be formally requested by contacting EVN Hanoi via email at evnhanoi@evnhanoi.vn. Additionally, Public socio-economic data from GSO (gso.gov.vn), such as income, unemployment, and CPI, complement the internal dataset.

Similarly, for electricity demand forecasting, internal operational data such as historical hourly load (MWh) comes from EVN's SCADA System Logs, also accessible by emailing evnhanoi@evnhanoi.vn. Complementary external data includes weather variables (temperature, humidity) from APIs like OpenWeatherMap and public holiday calendars from the Ministry of Labour of Vietnam, both freely available online.

b) Big Data, ML, and Ethical Considerations:

Big Data and ML enable timely, accurate prediction of customer defaults by leveraging high-volume, high-variety datasets with complex, high-dimensional features. These models can uncover hidden patterns across consumption, payment, and socio-economic variables, improving intervention targeting. However, ethical risks arise as customer privacy must be protected through strict anonymization, access controls, and a Data Protection Impact Assessment. Bias and fairness concerns, especially for low-income groups, require fairness audits and human-in-the-loop systems to ensure supportive, not punitive, outcomes. Governance is critical; a transparent framework with clear data use policies and informed consent ensures responsible, culturally sensitive deployment across diverse economic contexts.

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